Final Report
PK4 - Project
Topic: Checkers
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1. **Topic analysis**

Ultimately, tools such as the Unreal Engine and the C++20 language were used to create the project. All project assumptions have been implemented. This means the game is fully three-dimensional, relies on WAN network gameplay (which in short means it is possible to play with people from the other side of the world), and integration with the Steam platform was implemented to later save each player's progress to a database. The game is integrated with the Steam platform, which means running the Steam application is required in order to play.

Justification for the choices of classes, algorithms, and libraries

Libraries provided by the Unreal Engine were used. In some cases, it was not possible to use native C++ standard libraries, such as in the case of filesystem, where I had to use the IFileManager wrapper, which serves an identical purpose.

Most of the classes responsible for pawns (pieces) and physical elements such as the board inherit from the base AActor class. These classes override the necessary base class functions, or I add my own, such as pawn movement.

The classes responsible for win logic and matchmaking (searching for a game) inherit from the base AGameMode class. Similarly to the AActor class, I override existing functions from the base class or add my own.

2. External Specification

Intro (short film)





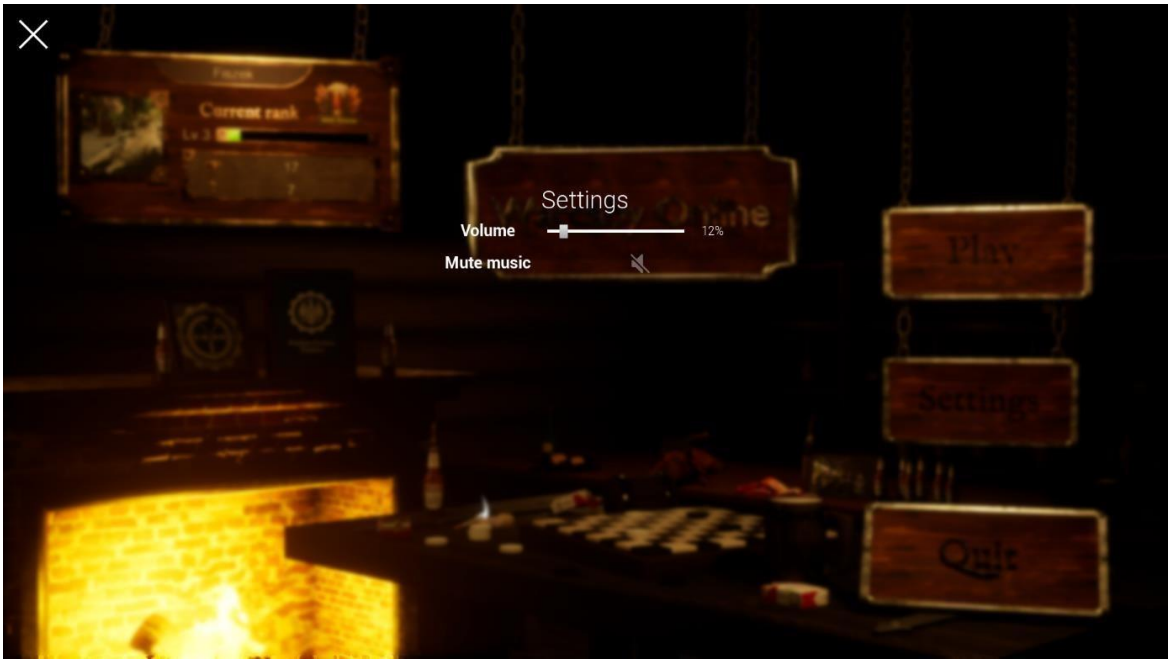
Main Menu

The menu was implemented fully in 3D. In addition to buttons, a tavern-themed environment was created so that the menu would not feel empty and dull. There is also a panel where saved progress can be viewed. Background music capturing the atmosphere of the scene plays in the menu as well.



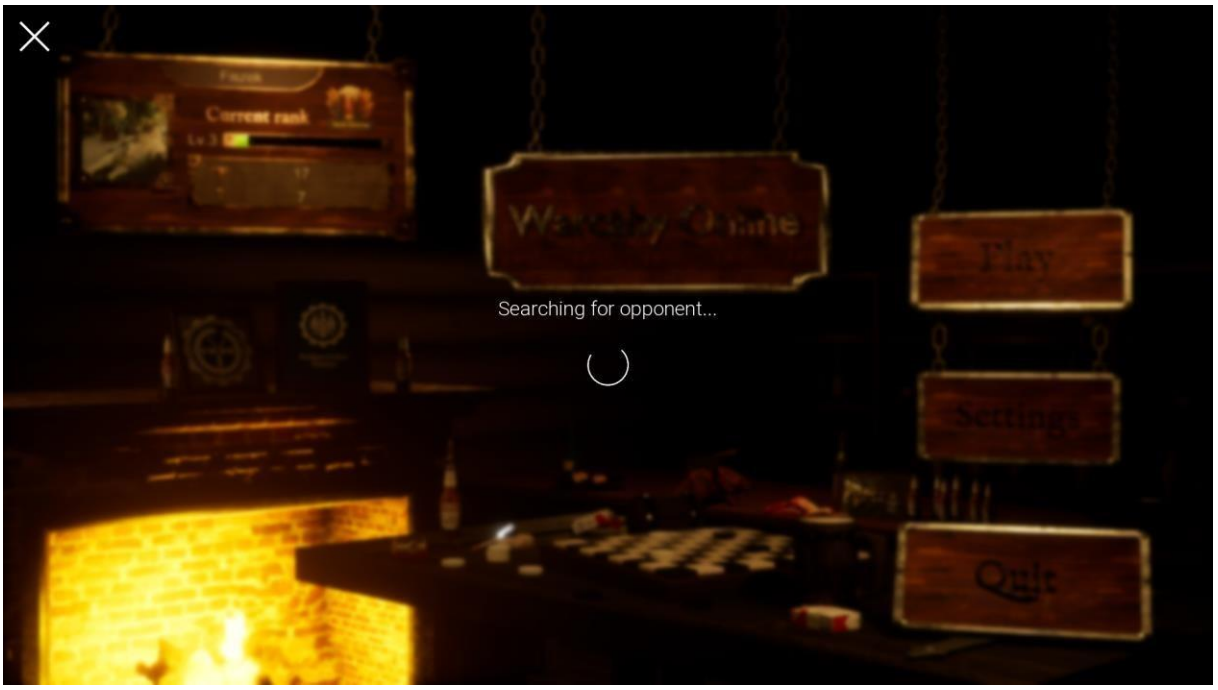
Settings

The settings include options such as volume control and the ability to completely mute the music. Pressing the 'X' button in the top-left corner saves the settings, which are then loaded from the disk upon restarting the game, so the player does not have to reconfigure them every time the application is launched.



Searching for opponent

A matchmaking system is implemented: pressing the 'Play' button in the menu brings up an opponent search screen. If no opponent is found, the player becomes the host and can be found by other players. Gameplay begins as soon as the host receives notification that there are two people in the lobby, at which point the game loads the chessboard level.



Gameplay

Rules: The entire match follows the classic rules of checkers:

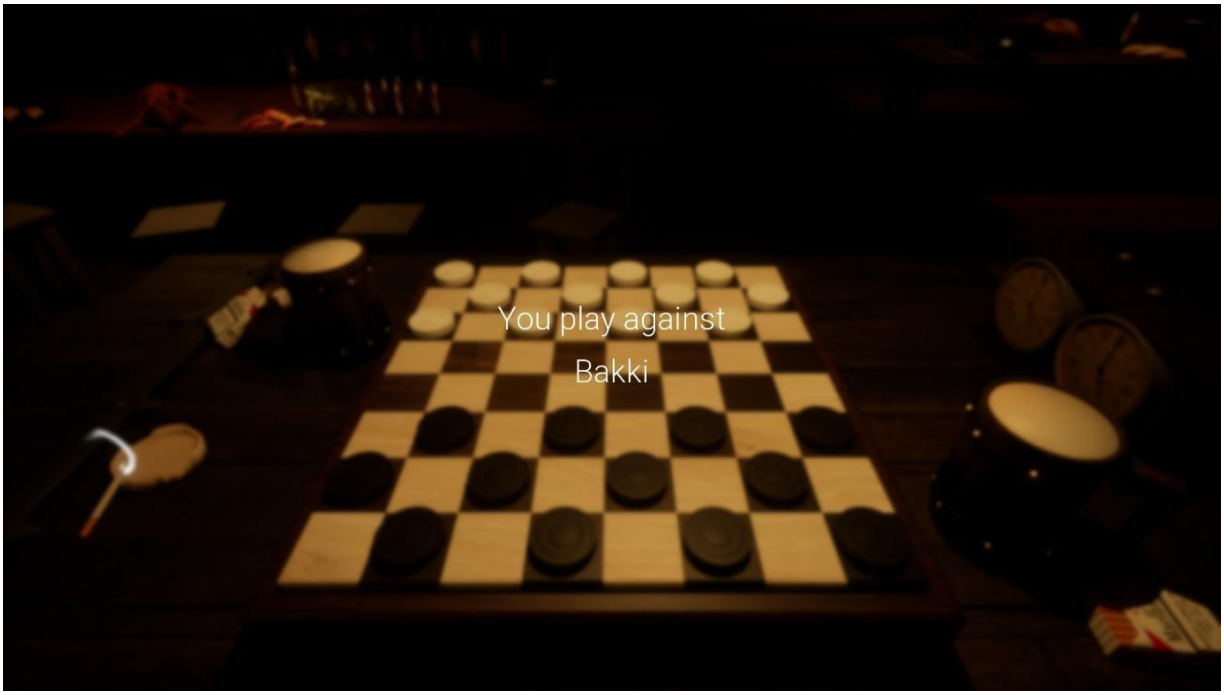
- Pawns can capture both forward and backward
- Captures are mandatory. If capture is possible, it must be executed.
- If a pawn reaches the opponents's back row, it is promoted to a king

After the board level loads, the player is informed who their opponent is (via a short text animation). Once the animation finishes, the white pieces begin the turn. Depending on whose turn it currently is, time elapses, which is visible on a timer located on the table. Each player has 5 minutes.

Gameplay can end in three ways: by surrender, running out of time, or capturing all of the opponent's pieces.

When the match ends, a result screen appears, and each player receives experience points and ranking points based on whether they won or lost the game.

Start of the match – presenting who the opponent is



Once the intro animation ends, gameplay begins. The timers start, the match is underway, and each player takes their turn to move their pieces.



Pawn promotion to a king



End of the match

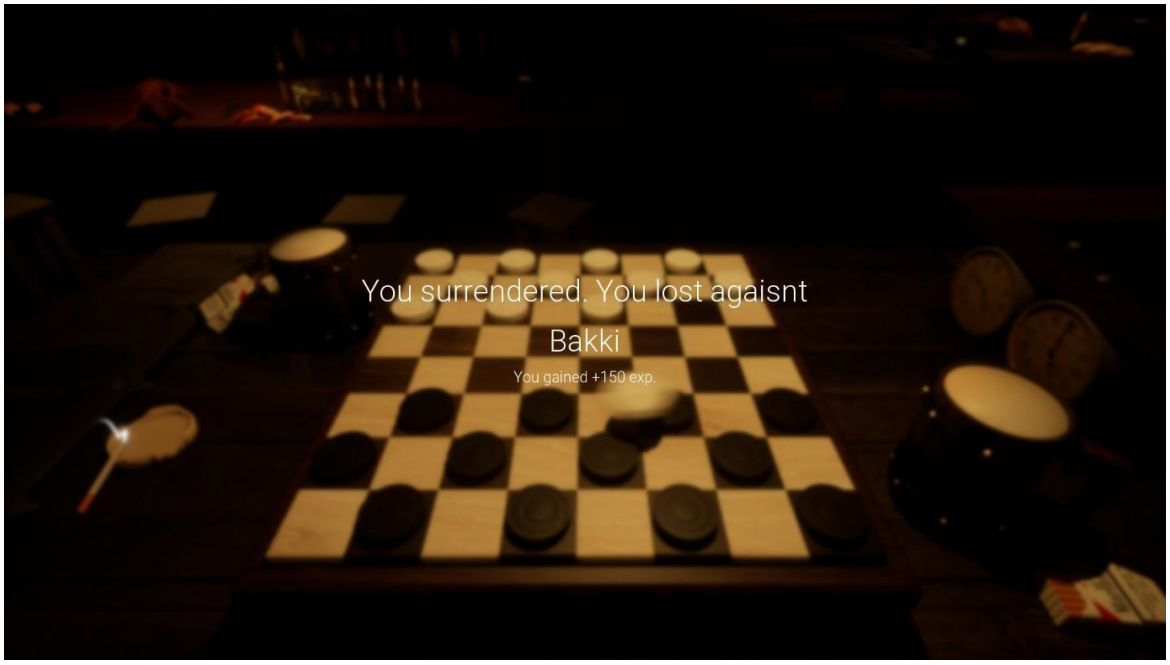
Several possible scenarios, including a won match



Lost match

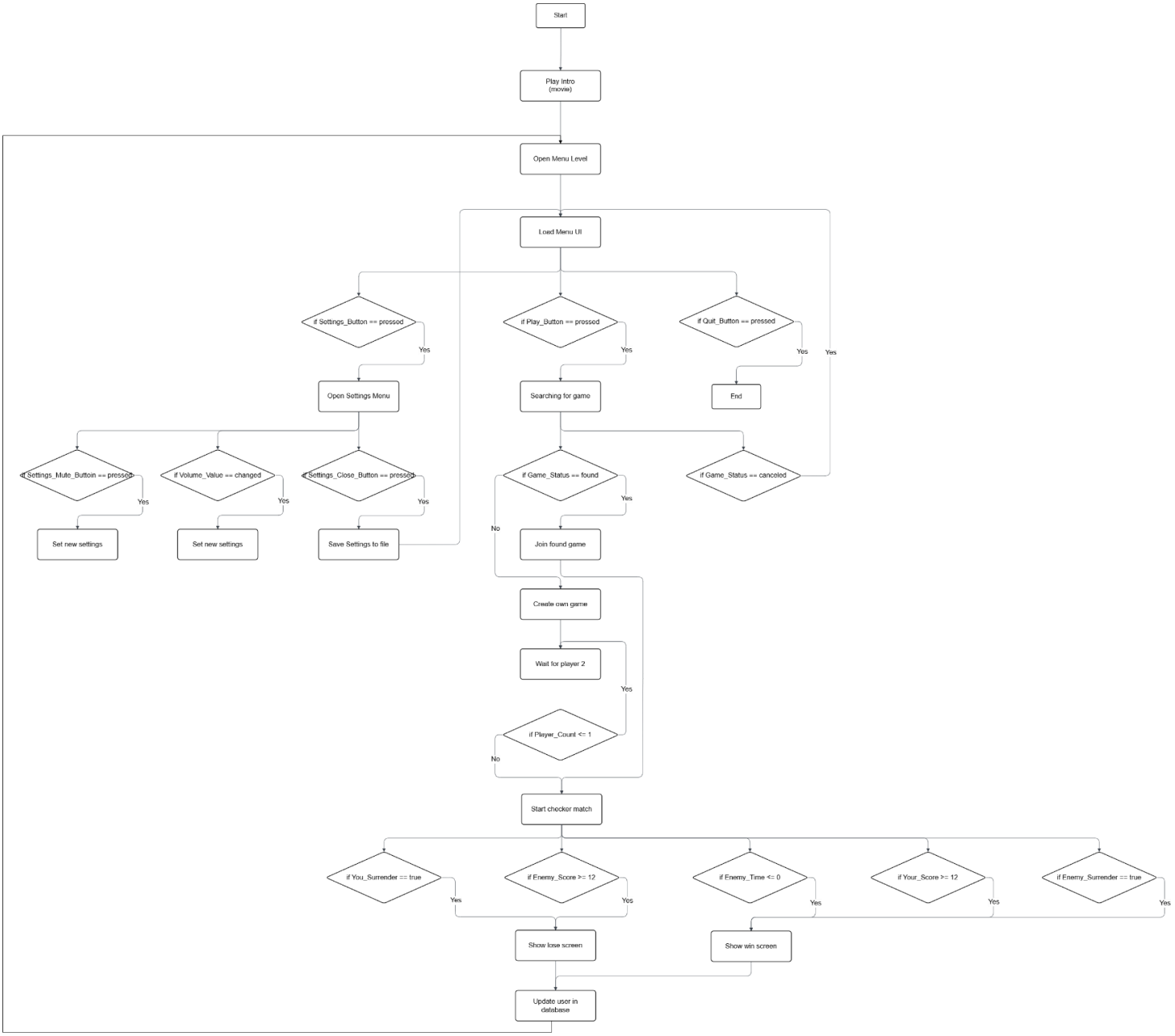


Surrender

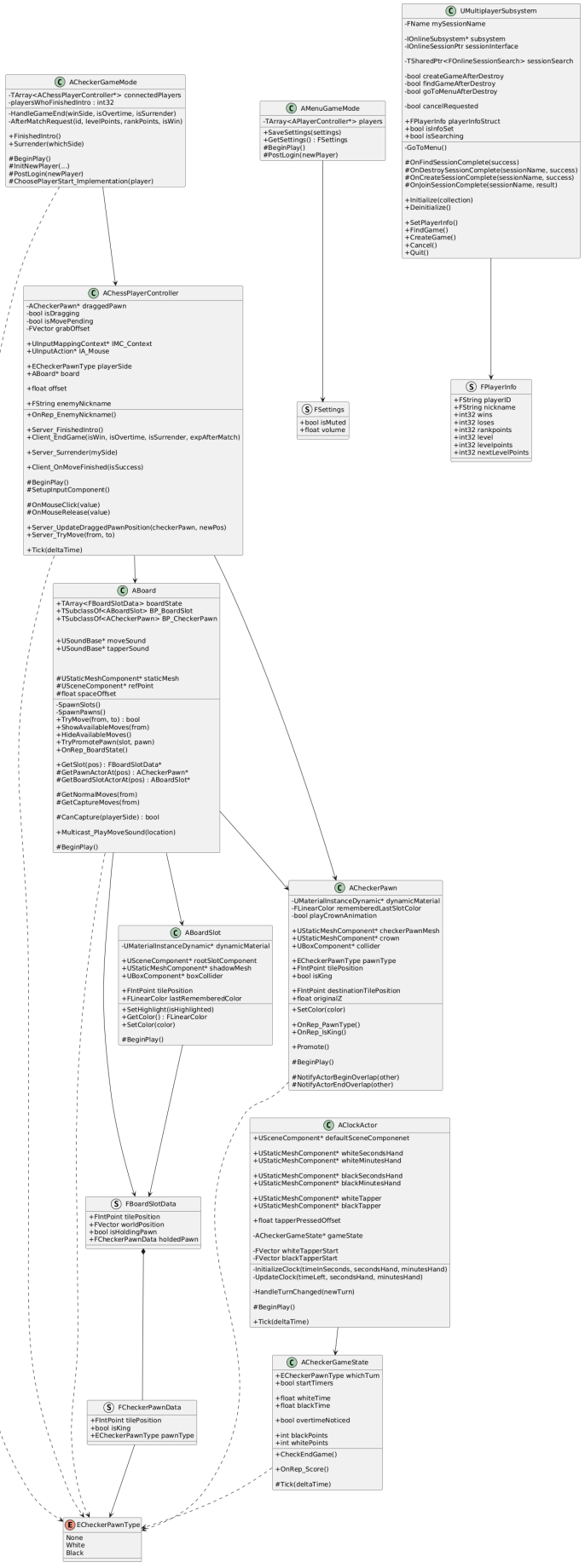


3. Internal Specification

UML Schematic



Class diagram



4. Testing and launching

Testing and launching the entire application, along with its operation, is described in **Section 2: External Specification**.

5. Remarks and Conclusions

Deciding to undertake this project allowed me to learn a new tool that I had no prior experience Unreal Engine and ultimately create my first game using it.

This task provided a clear understanding of the engine's workflow and its inner workings.

It was an engaging project that will certainly serve as a valuable addition to my portfolio.